



5 (a) (i) By writing $\cos\left(\frac{1}{12}\pi\right)$ as $\cos\left(\frac{1}{4}\pi - \frac{1}{6}\pi\right)$, show that $\cos\left(\frac{1}{12}\pi\right) = \frac{1}{4}(\sqrt{6} + \sqrt{2})$. [1]

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(ii) Show also that $\sin\left(\frac{1}{12}\pi\right) = \frac{1}{4}(\sqrt{6} - \sqrt{2})$. [1]

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The matrix $\mathbf{M}$ is such that $\mathbf{M} = \frac{1}{4} \begin{pmatrix} \sqrt{6} + \sqrt{2} & \sqrt{2} - \sqrt{6} \\ \sqrt{6} - \sqrt{2} & \sqrt{6} + \sqrt{2} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.

(b) The matrix $\mathbf{M}$ represents a sequence of two geometrical transformations in the x - y plane.

Give full details of each transformation, and make clear the order in which they are applied. [4]

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(c) Write $\mathbf{M}^{-1}$ as the product of two matrices, neither of which is $\mathbf{I}$.

[2]

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(d) Given that $y = mx$ is an invariant line of the transformation represented by $\mathbf{M}$, show that

$$m^2 \sin\left(\frac{1}{12}\pi\right) + 2m \cos\left(\frac{1}{12}\pi\right) - \sin\left(\frac{1}{12}\pi\right) = 0$$

and find the values of m in the form $a \cot\left(\frac{1}{12}\pi\right) + b \operatorname{cosec}\left(\frac{1}{12}\pi\right)$, where a and b are integers to be determined. [6]

[6]

[illegible]